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Comparison of Brain Structure Volumes in Insectivora and Primates.

IX. Trigeminal Complex

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With 4 Figures and 2 Tables

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Abstract and Summary: Volumes of the trigeminal complex (TR) were measured in 30 species of Insectivora, 3 species of Scandentia, 18 species of prosimians, 26 species of non-human simians and in man. The relative size showed a definite tendency to decrease from 159 in Insectivora to 56 in simians (expressed as size indices). The difference in the development of the TR between Insectivora and Primates is explained by differences in the role of their oro-facial region in exploratory behavior. The largest size was found in semiaquatic Insectivora (average 283). In semiaquatic forms, the reduction of the olfactory centers is compensated by an increase in TR size. The extremely long vibrissae innervated by the trigeminal nerve seem to have a teletactile function in detecting vibrations in the water produced by potential prey.

Key words: Morphometry, Trigeminal complex, Somatosensory system, Insectivora, Primates, Water-adaptation

Introduction

General somatosensory information from the oro-facial region to the brain is mediated by the trigeminal nerve. It is more important to the animal than information transmitted from other parts of the body for at least two reasons:

(1) The oro-facial region is the most complex area of the body surface, including such diversified areas as the mouth, with the tongue and the teeth; the nostrils; the cornea and the facial skin, which in certain mammals, most notably Insectivora, contains specialized tactile organs, the facial vibrissae. Sensory innervation of the meningeal lining of the anterior and middle cranial fossae, the muscles of mastication and of facial mimetic muscles is also mediated by the trigeminal system (DARIAN-SMITH 1973).

(2) The oro-facial region is the first part of the body to enter a new environment and thus provides the animal with precise information about changes in its immediate surroundings. The importance of trigeminal input, which supplements information obtained from other sensory organs, should therefore vary with the characteristics of the ecological niche.

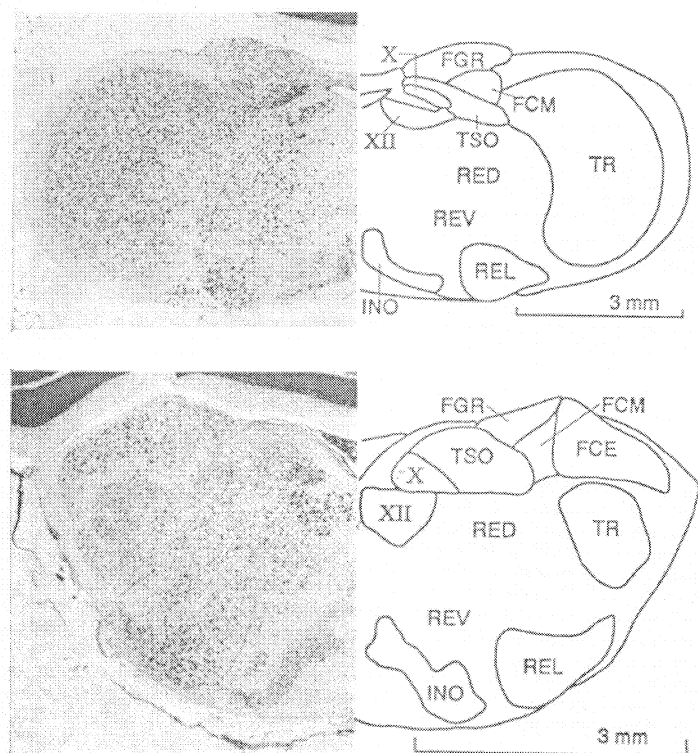
The present study was conducted primarily to compare changes in the relative size of the trigeminal complex (TR) in Insectivora and Primates in relation to their ecoethological adaptations and to contribute

to our understanding of size changes in various nuclei of the medulla oblongata, the most conservative structure of the brain (BARON et al. 1988).

Material and Methods

Volumes of the trigeminal complex (TR), including the nucleus sensorius principalis nervi trigemini (TRP) and the nucleus spinalis nervi trigemini (TRS), were measured in 78 species belonging to Insectivora (30 species), Scandentia (3 species) and Primates (18 species of prosimians and 27 species of simians) (Table 1). Detailed information on the methods of (1) fixation and preparation of the brains, (2) sections, (3) determination of the volumes and their conversion to fresh values, and (4) interspecific comparisons (percentage composition and allometric indices based on regression analyses) have been given in previous papers of this series (e. g., FRAHM et al. 1982; STEPHAN et al. 1982; BARON et al. 1983) and by STEPHAN et al. (1981). Body weights and the volumes of the total (net) brains and the major brain parts including the medulla were given in STEPHAN et al. (1981).

Allometric indices are numerical values of the differences of the various species from a reference line, the slope of which was determined by a canonical method (REMPE 1970; REMPE and WEBER 1972). The slope was found to be 0.59. In the 15 families and subfamilies of the Insectivora and Primates, in which, for our material, the body weight range of the species is above 2 (BoW-range > 2), the regression line slopes also converge at 0.59. A line with this slope was drawn through the average values for the subfamily of Tenrecinae; it has a y-intercept of -0.001 . Tenrecinae were chosen since their total brain and many of their brain structures



Figs. 1–2. Frontal sections through the medulla oblongata at the level of the sensorial trigeminal complex in *Potamogale velox* (Fig. 1) and *Galagoides demidoff* (Fig. 2). Cresylviolet stain.

FCE	nucleus fascicularis cuneatus externus
FCM	nucleus fascicularis cuneatus medialis
FGR	nucleus fascicularis gracilis
INO	complexus olivaris inferior
REL	nucleus reticularis lateralis
RED	nucleus reticularis dorsalis
REV	nucleus reticularis ventralis
TR	complexus sensorius nervi trigemini
TSO	nucleus tractus solitarii
X	nucleus dorsalis motorius nervi vagi
XII	nucleus nervi hypoglossi

have the smallest size indices of the Insectivora (and perhaps of all the extant eutherian mammals). Any value on the reference line (= line for average Tenrecinae = AvT) has an index of 1. The distances from the reference line are expressed as deviations from 1 and measure the extent to which the structural volumes deviate from those of typical Tenrecinae. The deviations, which are called size indices, are listed in their percentage form (i.e., $\times 100$) in Tables 1 and 2 and are scaled in Figure 4.

Results

Topography

The sensory trigeminal nuclear complex extends from the first cervical segment of the spinal cord up to the central gray of the mesencephalon. The complex is usually divided into three main parts: the nucleus mesencephalicus nervi trigemini (mesencephalic sensory nucleus, TRM), the nucleus sensorius principalis

nervi trigemini (main sensory nucleus, TRP) and the nucleus spinalis nervi trigemini (nucleus of the spinal tract, TRS). Although there are obvious differences in the relative extension of the various nuclei and subnuclei between Insectivora and Primates, the general topography, delimitation and structural characteristics are mutatis mutandis very similar in both orders.

The mesencephalic sensory nucleus (TRM) consists of a narrow band of scattered round cells at the lateral margin of the griseum centrale. This makes it impossible to measure its volume. Its size would better be determined by cell counts, which, however, have not yet been done. The two other main parts (TRP and TRS) could not be separated with sufficient accuracy and were, therefore, measured together under the abbreviation TR.

The most rostral portion of the TR is the TRP. In Insectivora its rostral tip reaches a level rostral to the superior olivary complex. In Primates, the rostral pole lies more caudally, but it is still rostral to the caudal end of the medial parabrachial nucleus. Caudally, it merges with the rostral part of the subnucleus oralis of TRS.

The supratrigeminal nucleus mentioned by LORENTE DE NÓ (1922) (ex DARIAN-SMITH 1973) is here considered a part of TRP and, hence, is included in the TR volume. It is located at the dorsomedial angle of TRP at the level of the motor trigeminal nucleus.

The TRS is continuous with TRP, rostrally, and the dorsal horn of the spinal cord, caudally. The nucleus may be subdivided into three continuous regions or subnuclei which differ functionally and cytoarchitecturally, although the precise limits are not obvious in Nissl stain: subnuclei oralis, interpolaris and caudalis (OLSZEWSKI 1950).

Percentage composition relative to the oblongata

The TR, representing on the average 10.2% ($N = 78$), is, apart from the reticular formation, the largest griseum of the medulla oblongata. The percentage distinctly decreases from an average of 18.1% in Insectivora to 6.6% in Scandentia and 6.0% in prosimians (Table 2). In nonhuman simians, TR occupies 4.5% of the oblongata, a figure slightly smaller than that obtained in prosimians. In Insectivora, the percentage varies from 11.5% in *Hylomys suillus* to 24.6% in *Sorex cinereus*; in prosimians, from 4.0% in *Lemur albigrons* to 8.1% in *Perodicticus potto*; and in simians, from 3.4% in *Pan troglodytes* and *Erythrocebus patas* to 5.9% in *Cebuella pygmaea*. With 4.2%, *Homo* has one of the smallest TRs relative to the oblongata.

In insectivoran families, the variability of the percentage size of TR relative to the oblongata may be

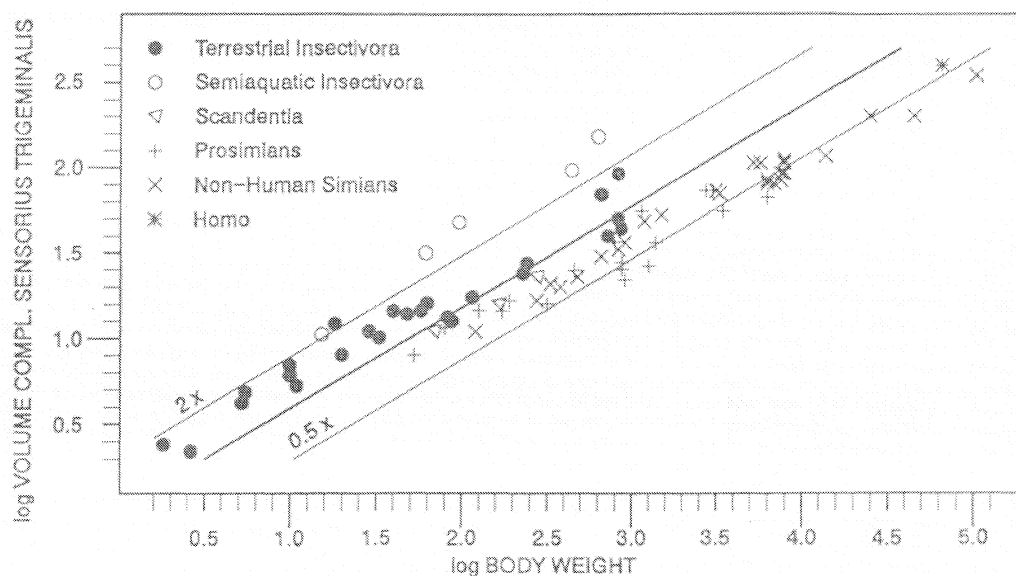


Fig. 3. Volumes (in mm^3) of trigeminal complex (TR) plotted against body weight (in grams) in a double logarithmic scale. The thick reference line is through the average of the 4 Tenrecinae (AvT line; see Material and Methods) and has the equation

$$\log \text{TR volume} = -0.001 + 0.59 \times \log \text{body weight}.$$

The thinner parallel lines are in multiples of 2 (above 2 times, below 1/2 time) and refer to the size indices scaled in Fig. 4 and given in Tables 1–2 in their percentage form ($\times 100$).

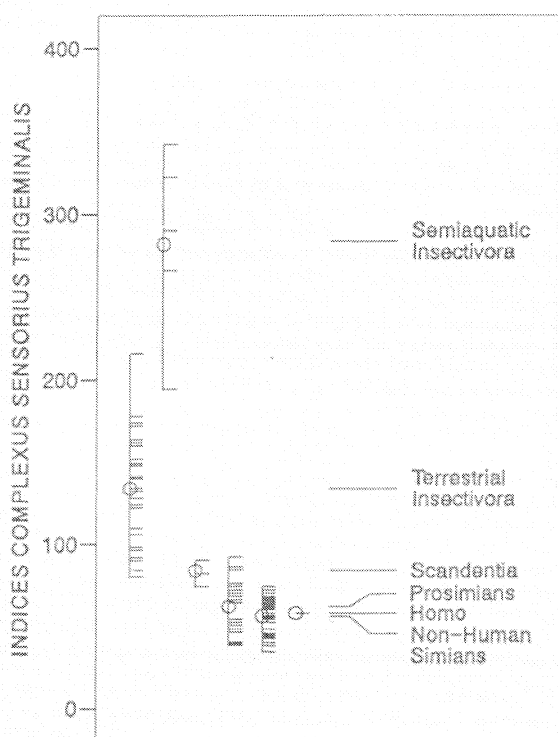


Fig. 4. Indices of the trigeminal complex (TR) in various taxonomic units and ecoethological types. The indices are numerical values of the distances from the thick reference baseline given in Fig. 3. Each of the small horizontal bars represents the index for a species. On the left, the ranges (vertical bars) and the means (o) are given, on the right, the sequence of the means is scaled and the names of the units are given.

very high. In Tenrecidae, the percentages vary from 14.1% in *Echinops* to 24.1% in *Micropotamogale*. The average in the Tenrecinae is 15.0%, whereas that in the Potamogalinae is 22.5%. In Talpidae, the TR in *Talpa europaea* (Talpinae) occupies 13.9% of the oblongata, but in the Desmaninae, it averages 23.0%.

Although percentage values indicate certain trends in size variation of the TR they may represent spurious results because they depend not only on the size of the structure under comparison but also, to a considerable extent, on the size of the other components included in the reference system.

Size indices

The size indices of TR in the Insectivora range from 81 ($= 0.81 \times \text{AvT}$) in *Atelerix algirus* to 342 ($= 3.42 \times \text{AvT}$) in *Potamogale velox*. In *P. velox*, the relative size was about 4.2 times larger than in an isoponderous *Atelerix*. In fact, *P. velox* had the largest TR of all the species so far investigated. A similarly high index was found in *Micropotamogale ruwenzorii*. Low indices were obtained in all Erinaceinae (mean: 87), in *Talpa europaea* (99) and in Tenrecinae (mean: 100). One of the most intriguing results was the great interspecific variation within several families. The coefficient of variation (CV) of the Tenrecidae, which have a mean index of 172, was 64.3%. Besides the Tenrecinae, the reference group, *Microgale talazaci*, the only representative of the Oryzoricinae, had an

Table 1. Volumes, percentages and indices of the trigeminal complex in Insectivora, Scandentia and Primates. The indices are distances from the Tenrecinae baseline (AvT) given in their percentage form ($= \times 100$).

* = semiaquatic species

	Vol.	% of OBL	Index		Vol.	% of OBL	Index
INSECTIVORA				PROSIMIANS			
<i>Solenodon paradoxus</i>	69.2	14.7	149	<i>Lepilemur ruficaudatus</i>	22.2	5.0	40
<i>Tenrec ecaudatus</i>	49.3	15.6	92	<i>Avahi l. laniger</i>	26.3	4.8	39
<i>Setifer setosus</i>	27.5	15.2	110	<i>A. l. occidentalis</i>	25.9	5.1	48
<i>Hemicentetes semispinosus</i>	17.6	15.1	107	<i>Propithecus verreauxi</i>	56.0	4.6	46
<i>Echinops telfairi</i>	12.7	14.1	91	<i>Indri indri</i>	65.6	4.9	38
<i>Microgale talasaci</i>	13.8	16.4	140	<i>Daubentonia madagasc.</i>	71.9	4.7	67
<i>Potamogale velox*</i>	151.4	21.0	342	<i>Loris tardigradus</i>	15.6	6.7	52
<i>Micropotam. ruwenzorii*</i>	47.5	24.1	322	<i>Nycticebus coucang</i>	36.6	6.9	71
<i>Chrysochloris stuhlmanni</i>	14.4	21.1	163	<i>Perodicticus potto</i>	54.8	8.1	86
<i>Echinosorex gymnurus</i>	90.5	14.6	173	<i>Galago senegalensis</i>	16.7	6.6	77
<i>Hylomys suillus</i>	14.5	11.5	133	<i>Otolemur crassicaudatus</i>	28.1	5.2	53
<i>Atelerix algirus</i>	39.9	11.6	81	<i>Galagoides demidoff</i>	12.2	7.2	92
<i>Erinaceus europaeus</i>	44.8	13.4	84	<i>Tarsius spp.</i>	14.4	7.0	84
<i>Hemiechinus auritus</i>	23.9	12.9	96				
<i>Sorex araneus</i>	5.99	20.6	153	SIMIANS			
<i>S. cinereus</i>	4.22	24.6	161	<i>Callithrix jacchus</i>	16.3	5.1	59
<i>Microsorex hoyi</i>	2.15	19.7	122	<i>Cebuella pygmaea</i>	10.8	5.9	64
<i>Neomys fodiens*</i>	9.66	21.3	194	<i>Saguinus midas tamarin</i>	20.8	4.9	67
<i>Blarina brevicauda</i>	8.09	18.8	140	<i>S. oedipus</i>	20.0	4.9	60
<i>Crocidura flavescens</i>	10.8	18.0	148	<i>Callimico goeldii</i>	23.2	5.0	61
<i>C. hildegardeae</i>	6.81	21.7	175	<i>Cebus albifrons</i>	74.2	4.3	65
<i>C. russula</i>	5.29	18.5	128	<i>Aotus trivirgatus</i>	33.5	4.9	64
<i>Suncus etruscus</i>	2.33	21.0	162	<i>Callicebus moloch</i>	36.2	4.6	66
<i>S. murinus</i>	9.80	18.0	123	<i>Saimiri sciureus</i>	29.9	4.2	65
<i>Sylvisorex megalura</i>	4.88	18.2	179	<i>Pithecia monacha</i>	53.9	5.3	72
<i>Ruwenzorisorex suncooides</i>	12.0	21.9	217	<i>Alouatta seniculus</i>	79.1	5.0	45
<i>Scutisorex somereni</i>	15.5	18.7	134	<i>Ateles geoffroyi</i>	89.2	4.9	45
<i>Desmana moschata*</i>	97.0	24.1	267	<i>Lagothrix lagotricha</i>	107.0	5.1	69
<i>Galemys pyrenaicus*</i>	32.3	22.0	290	<i>Macaca mulatta</i>	107.5	5.4	54
<i>Talpa europaea</i>	13.3	13.9	99	<i>Cercopithecus albigena</i>	108.7	4.1	55
SCANDENTIA				<i>Papio anubis</i>	205.3	3.9	52
<i>Tupaia glis</i>	15.5	7.1	75	<i>Cercopithecus ascanius</i>	68.0	4.2	56
<i>T. minor</i>	11.1	6.4	91	<i>C. mitis</i>	84.5	4.2	49
<i>Urogale everetti</i>	22.5	6.3	82	<i>Miopithecus talapoin</i>	48.1	4.7	74
PROSIMIANS				<i>Erythrocebus patas</i>	89.5	3.4	45
<i>Microcebus murinus</i>	7.84	7.9	75	<i>Colobus badius</i>	78.2	3.9	42
<i>Cheirogaleus major</i>	25.6	6.7	70	<i>Pygathrix nemaeus</i>	82.3	3.7	43
<i>C. medius</i>	14.3	7.1	67	<i>Nasalis larvatus</i>	113.7	3.9	41
<i>Lemur albifrons</i>	36.6	4.0	51	<i>Hylobates lar</i>	103.2	4.6	63
<i>Varecia variegatus</i>	71.6	5.0	64	<i>Pan troglodytes</i>	199.6	3.4	35
				<i>Gorilla gorilla</i>	353.0	4.7	39
				<i>Homo sapiens</i>	400.9	4.2	58

index of 140 and the Potamogalinae ($N = 2$), a mean index of 332. The CV of the Erinaceidae was 34.6% and that of the Talpidae 47.7%. The mean index for the Echinosoricinae (153) was nearly twice as large as that for the Erinaceinae (87). In the Talpidae, the mean index for the Desmaninae (279) was 2.8 times larger than the index for *Talpa europaea*. The lowest coefficient of variation was found in the Soricidae ($CV = 20.5\%$). Among the Soricinae (mean: 154), *Neomys fodiens* had the highest index.

The mean index of Scandentia (83) was low compared with Insectivora. It was slightly lower than that of the Erinaceinae, which had the lowest mean index of Insectivora.

In all prosimians, the TR was clearly reduced compared with the reference group, the Tenrecinae. The mean index (62) was less than 40% of that found in Insectivora. The lowest indices were obtained in *Lepilemur ruficaudatus* (40) and the Indridae (mean: 43). *Galagoides demidoff* (92), *Perodicticus potto* (86)

Table 2. Means of percentages and indices in various ecoethological and taxonomic groups

	N	% of OBL	Index		N	% of OBL	Index
INSECTIVORA	30	18.1	159	PROSIMIANS	18	6.0	62
Terrestrial	25	17.2	134	Cheirogaleidae	3	7.2	71
Semiaquatic	5	22.5	283	Lemurinae	2	4.5	57
Solenodontidae	1	14.7	149	Lepilemurinae	1	5.0	40
Tenrecinae	4	15.0	100	Indriidae	4	4.8	43
Oryzorictinae	1	16.4	140	Daubentonidae	1	4.7	67
Potamogalinae	2	22.5	332	Lorisinae	3	7.2	70
Chrysochloridae	1	21.1	163	Galaginae	3	6.3	74
Echinosoricinae	2	13.0	153	Tarsiidae	1	7.0	84
Erinaceinae	3	12.6	87	SIMIANS	27	4.5	56
Soricinae	5	21.0	154	Callitrichidae	5	5.1	62
Crocodyrinae	8	19.5	158	Cebidae	8	4.8	61
Talpinae	1	13.9	99	Cercopithecinae	7	4.3	55
Desmaninae	2	23.0	279	Colobinae	3	3.8	42
SCANDENTIA	3	6.6	83	Pongidae	3	4.2	46
				Hominidae	1	4.2	58
				OVERALL	78	10.2	98

N = number of species

and *Tarsius* spp. (84) had the best developed TR of all the prosimians and were in the range of variation of Tenrecinae and Erinaceinae.

The mean index of simians (56) was only slightly smaller than that of prosimians, indicating that there was no clear trend for a decrease in the TR from prosimians to simians. The lowest indices were similar in the two groups. In fact, *Pan troglodytes* (35) had only a slightly smaller index than *Indri indri* (38). The highest index among simians was found in *Miopithecus talapoin* (74). Both families of New World monkeys, the Callitrichidae and Cebidae, had very similar mean indices i.e., 62.2 and 61.4, respectively. The CV for the Callitrichidae (5.3%) was exceptionally low, that of Cebidae relatively high (17%). In fact, *Pithecia monacha* had one of the highest indices in the simians (72), whereas *Alouatta seniculus* and *Ateles geoffroyi* had indices (45) far below the simian average. In Old World monkeys, the mean index found in the Cercopithecinae (55) was considerably higher than that found in the Colobinae (42). The Pongidae, with a mean index of 45.7, had the highest CV (33%) of all the simians. The indices of *Pan troglodytes* and *Gorilla gorilla* were among the lowest of all species so far investigated. The index of *Hylobates lar*, on the other hand, was above the simian average. Finally, the index of *Homo* was close to the mean index for simians.

Discussion

The most striking finding of this study is the decrease in the TR size indices from Insectivora to Primates and their great range of variation, particularly in Insectivora. Ethological and physiological studies

have shown that the trigeminal system plays an important role in orientation and exploration, as well as in feeding behavior.

Somatosensory input conveyed from the mechanoreceptors, thermoreceptors and nociceptors of the orofacial region to the TR provide the animal with important information as it enters and explores new environments (DARIAN-SMITH 1973). With the development of vibrissae in many mammals, the animal cannot only actively explore the environment well beyond its facial profile, but may also be capable of effective telereception, particularly in an aquatic environment. Experiments with rats involving section of the infraorbital nerve, which innervates the vibrissae, have shown severe behavioral impairment due to the loss of somatosensory information. Although the denervated animal remains active, there are noticeable differences in its gait and posture. While the normal animal keeps its head above the substrate, which is contacted only by the vibrissae, the animal with denervated vibrissae has the snout in direct contact with the substrate and it actually bumps into objects placed in its path (ZEIGLER 1983).

Another important function of the trigeminal system is its contribution to the neural control of feeding, particularly in grasping and mastication of food. Information about the characteristics of the objects in the mouth appears to be essential for proper mastication. Behavioral experiments with a trigeminally deafferented rat have shown that the lack of trigeminal sensory input produces a profound disruption of ingestive behavior (ZEIGLER 1983).

Insectivora: The differences, between Insectivora and Primates in the development of the TR, can be at

least partly explained by differences in the role of the oro-facial region in exploratory behavior. Facial whiskers are very prominent in all Insectivora, although there may be differences in the basic pattern and size. EISENBERG and GOULD (1970) describe mystacial, genal, superorbital and mental vibrissae in the Tenrecinae, and GRASSÉ (1955), in *Erinaceus*. HUTTERER (1985), however, mentions only mystacial and submental vibrissae in shrews. There is no positive correlation between the type of whiskers and the size of the trigeminal complex. Although they seem to have fewer types of whiskers, shrews have higher size indices than the Tenrecinae and Erinaceinae. The salient feature common to all Insectivora is the ensemble of the mystacial vibrissae, which are complex tactile exploratory organs, capable of active, rapid and accurate movement in space (DARIAN-SMITH 1973).

The best developed TR by far was found in semiaquatic Insectivora (mean index: 283). Consistent with the huge size of TR, the vibrissae attain the greatest development in semiaquatic species. In addition to the very abundant and long mystacial vibrissae arranged in 11 to 13 lines, many other facial tactile hairs have been described (GUTH et al. 1959; VERHEYEN 1961a, b; KUHN 1964). The proboscis of the water moles, such as *Galemys*, contains two types of touch organs. The proximal part is covered on each side with 80 to 100 vibrissae (BAUCHOT et al. 1973). Around the nostrils are Eimer's organs (ARGAUD 1944; RICHARD and VIALARD 1969; BAUCHOT et al. 1973), as many as about 100,000 over a few mm² (RICHARD 1985). Previous studies (STEPHAN and BAUCHOT 1959; STEPHAN and SPATZ 1962; BAUCHOT and STEPHAN 1968) have already suggested that the long whiskers in *Galemys* and other semiaquatic Insectivora may have a teletactile function.

In the water shrew *Neomys*, the least adapted semiaquatic form of the Insectivora in this study, the morphologic modifications for aquatic life are less pronounced than in the other semiaquatic species. Consistent with these morphologic characteristics, *Neomys* has the least developed TR of all the semiaquatic Insectivora.

The TR size index of the terrestrial forms is very variable even within the same family. Among Tenrecidae *Microgale* (140) has a better developed TR than Tenrecinae. The shrew-like *Microgale* in tropical Madagascar occupies an ecological niche similar to that of shrews in the temperate boreal zone, with which it shares a similar TR size index. *Microgale* has been observed inhabiting moist meadows covered with dense grass and reeds. The vegetation covering their runways was so dense and matted that sunlight did not penetrate to the ground (WALKER 1975). It is

reasonable to suspect that they rely much more on tactile information than the mainly surface-dwelling Tenrecinae. While all Erinaceidae are terrestrial forms, there are significant differences in the size indices between the Echinosoricinae (gymnures and moonrats) and Erinaceinae (hedgehogs) (153 versus 87). The two groups differ in their morphological and ecoethological adaptations. Although we lack detailed information on the exploratory behavior of gymnures, there are indications that the complexity of their environment may be greater than in hedgehogs. Gymnures live in humid forests with dense undergrowth. *Echinosorex* may be aquatic to some extent and *Hylomys* has been observed to climb in trees (HERTER 1938, 1979; LIM 1967; WALKER 1975; GOULD 1978). True hedgehogs avoid humid regions.

Although the Soricinae and Crocidurinae differ in biotope and feeding ecology, there are no clear diverging trends in TR development. The Soricinae generally live in a densely covered biotope and spend a considerable amount of their active time in underground tunnels (HUTTERER 1976). The Crocidurinae are predominantly surface dwellers. However, *Sylvisorex megalura* and particularly *Ruwenzorisorex suncooides* have higher indices than the other terrestrial Soricidae. *Sylvisorex megalura* is scansorial. Since it was the only scansorial species in this study, no statement can be made about the relation between increased TR and climbing abilities. The TR index of *Ruwenzorisorex suncooides* falls in the range of variation of semiaquatic species. It is interesting to note that the main olfactory bulb is reduced in *Ruwenzorisorex suncooides* as in semiaquatic species.

The two fossorial species of this study differ in their TR development. While the index for *Chrysochloris stuhlmanni* was above the mean for the terrestrial Insectivora, that for *Talpa europaea*, surprisingly, was far below, although its whiskers and Eimer's organs (EIMER 1871; BIELSCHOWSKY 1907; SLONAKER 1920; QUILLIAM 1966) are well developed and tactile organs seem to play an important role in orientation in subterranean tunnels (KRISZAT 1940a, b; GODET 1951; QUILLIAM and ARMSTRONG 1963). However, the hands as well as the snout are sensitive tactile organs (KRISZAT 1940a, b; QUILLIAM 1966). Furthermore, the entire fur plays an important role in tactile sensation. Moles also have Pinkus plates, small skin elevations containing several Merkel disk type receptors, especially in the abdominal region (QUILLIAM 1966). It is therefore possible that *Talpa* relies less on the oro-facial region for tactile information.

Primates: The exploratory function of the oro-facial region becomes much less important in Primates as the hand evolves and assumes this role (DARIAN-SMITH 1973). Contrary to Insectivora, Primates gen-

erally have less developed vibrissae or lack them completely (GRASSÉ 1955). Because they are arboreal or walk, keeping their head far above the ground, Primates rely more on vision than on general somatosensory input from the oro-facial region.

All prosimians are known to have vibrissae. Differences in the development of TR are related to feeding ecology. Folivorous species have the least developed TR. Their mean index is 46.6 as compared to 73.5 in insectivorous and 70.6 in frugivorous species. Most genera of the New World monkeys have facial vibrissae (GRASSÉ 1955), but they are, in general, less conspicuous than in prosimians. As in prosimians, size differences are mainly paralleled by the trophic niche. The folivorous *Alouatta* has the lowest index of the New World monkeys. The lack of functionally efficient vibrissae in Old World monkeys seems to have little effect on the TR indices (mean: 51) in comparison to New World monkeys (mean: 62). Here again, the size of the nucleus is related to the trophic niche. The folivorous Colobinae have a smaller TR (mean: 42) than the Cercopithecinae.

The functional significance of reduced TR size in folivores is not immediately obvious. The reduction may be a consequence of the decrease in the allometric brain size often found in folivores. However, comparative data suggest that size indices for TR do not follow the same trend as those for the total brain or the medulla oblongata. The size index for the medulla of strict folivores is 79% of the index in insectivorous prosimians, the TR only 57%. In generalist folivores (*Lemur* and *Varecia*), the medulla is 1.15 times larger than in insectivorous and 1.21 times larger than in frugivorous prosimians. In contrast, the TR index of generalist folivores is only 81% of that of frugivorous and 78% of that of insectivorous species. In simians, the reduction of the TR is also more pronounced than that of the medulla oblongata. The size index of the medulla in the folivorous Colobinae is 84% of that of the Cercopithecinae, whereas the TR index reaches only 76% of the index for the same structure in the Cercopithecinae.

Although in some species, particularly in diurnal prosimians, the reductions of oro-facial sensibility may be compensated by better developed vision (FRAHM et al. 1984), there is no obvious compensation either by olfaction or by vision in the other folivorous species (BARON et al. 1983; FRAHM et al. 1984). Based on the discussions in previous papers (e.g., BARON et al. 1983; FRAHM et al. 1984; STEPHAN et al. 1988), one may assume that differences in size indices may be related to differences in niche complexity. In fact, the ubiquity of food in the case of folivorous species renders the food supply more predictable and, hence, their niche less complex.

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